WORKSHEET

Chapter 6: Displacement, velocity and acceleration

1 The velocity of a particle is given by $v = 4 - 3t$ cms^{-1}. If the particle is 5 cm from the origin after 2 s, find the displacement after 7 s.

2 A particle has a constant acceleration of 12 ms^{-2}. If the particle has a velocity of 2 ms^{-1} and is 3 m from the origin after 5 s, find its displacement after 10 s.

3 A projectile is accelerating at a constant rate of -9.8 ms^{-2}. If it is initially 2 m high and has a velocity of 4 ms^{-1}, find the equation for the height of the projectile.

4 The velocity of a particle is given by $v = \dfrac{t}{t^2 + 3}$ ms^{-1}. If the particle is initially at the origin, find its displacement after 10 s, correct to 2 decimal places.

5 The acceleration of an object is given by the expression $a = 3 \sin\left(2t + \dfrac{\pi}{4}\right)$ ms^{-2}. Determine an expression for the velocity of the object and find its velocity at $t = 3$ s.

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6 The velocity of a particle at time t seconds is given by $v = 2t^2 - \sqrt{t+1}$ ms^{-1}. Initially, the position of the particle is 5 m from the origin. What is the position of the particle 4 s later?

7 The acceleration of a particle is $a = e^{2t} - 5$ ms^{-2}. If the particle starts from rest (i.e. $v = 0$ when $t = 0$), how far is the particle from its starting point 2 s later?

8 The acceleration of a particle is given by $a = \cos(\pi t)$ ms^{-2}. The initial velocity of the particle is $\frac{1}{2\pi}$ ms^{-1}.

a Find the displacement of the particle after 1.5 s.

b Find the total distance travelled by the particle in the first 1.5 s.

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Answers

1 -42.5 cm

2 163 m

3 $h = -4.9t^2 + 4t + 2$

4 1.77 m

5 $v = -1.5\cos\left(2t + \frac{\pi}{4}\right)$; About -1.3148 ms^{-1}.

6 $\frac{157}{3} - \frac{10\sqrt{5}}{3}$ m

7 About 2.4 m

8 **a** About 0.34 m

b About 0.409 m